

MUHAMMAD HAMZA TARIQ

Full-Stack Developer | IoT & Real-Time Systems | Angular | Node.js | React

Islamabad, Pakistan · +92 318 335 5301 · hamzatariqno1@yahoo.com · [linkedin.com/in/hatariq1](https://www.linkedin.com/in/hatariq1) · github.com/i190605

PROFESSIONAL SUMMARY

Results-driven Full-Stack Developer with 2+ years of hands-on experience architecting and delivering end-to-end web applications and IoT-integrated platforms. Proven expertise in Angular, React.js, Node.js/Express, RESTful API design, and real-time data visualization. Adept at working across the full SDLC — from requirements gathering and system design through deployment and post-launch support — in Agile/Scrum environments. Passionate about clean, maintainable code, performance optimization, and collaborating with cross-functional teams to ship high-impact products at scale.

PROFESSIONAL EXPERIENCE

Full-Stack Developer

Tech Access | Deployment Partner: Visionvalley · May 2025 – Present · Islamabad, Pakistan

- Engineered a production-grade Smart Irrigation IoT platform end-to-end, integrating sensor data pipelines, REST API endpoints, and real-time dashboards — reducing manual monitoring overhead by an estimated 40%.
- Developed and maintained full-stack web applications with Angular 17 (frontend) and Node.js/Express (backend), implementing comprehensive CRUD operations and role-based access control for multi-tenant data management.
- Integrated IoT devices (sensors, actuators) with web applications via the Cumulocity IoT platform, enabling real-time monitoring, remote control, and alerting of connected irrigation systems.
- Built and optimized scalable RESTful APIs to handle high-frequency IoT device telemetry, improving data processing throughput and reducing API response latency.
- Implemented interactive data visualization dashboards using Chart.js and D3.js to surface key irrigation metrics, enabling data-driven decision-making for clients.
- Authored comprehensive API documentation (Swagger/OpenAPI) and technical runbooks, accelerating onboarding and reducing support escalations.

Frontend Developer

Code-IO · Aug 2023 – Nov 2023 · Islamabad, Pakistan · Hybrid

- Developed responsive, accessible UIs using React.js and modern JavaScript (ES6+), contributing to a 20% increase in user engagement metrics within 3 months.
- Collaborated with backend engineers to integrate RESTful APIs, implementing async data fetching, error handling, and state management patterns (Redux/Context API).
- Built and maintained a library of reusable, scalable React components, reducing UI development time by promoting consistency across the application.
- Participated in code reviews and Agile sprint ceremonies, consistently delivering features on schedule within a fast-paced team environment.

Information Technology Intern

FFBL (Fauji Fertilizer Bin Qasim Limited) · Jul 2022 – Aug 2022 · Karachi, Pakistan

- Supported enterprise IT operations, contributing to the development and testing of internal tools and workflows in a large-scale manufacturing environment.
- Delivered technical end-user support across departments, troubleshooting hardware, software, and network issues with a focus on minimal downtime.

KEY PROJECTS

Smart Irrigation System IoT Platform

Stack: Angular · Node.js · Cumulocity IoT · MQTT · REST API · Chart.js

- Architected an end-to-end IoT irrigation platform with real-time sensor dashboards, Cumulocity device management, and CRUD modules for system configuration.
- Integrated MQTT-based device telemetry pipelines with a Node.js backend, enabling live monitoring and remote actuation of irrigation hardware across field deployments.

- Delivered deployment documentation and post-launch support runbooks for cross-functional rollout teams serving agricultural and urban client sites.

Animal Feed ERP System *Full-Stack ERP Platform*

Stack: React (Vite) · Node.js · Express.js · JWT · REST APIs · RBAC · Git/GitHub

- Architected and delivered a modular full-stack ERP covering procurement, production, inventory, transport, sales, reporting, dashboards, and audit workflows for animal feed operations.
- Built secure REST APIs with JWT authentication and role-based authorization (RBAC), ensuring granular access control across all business modules.
- Implemented role-aware frontend journeys in React (Vite), establishing clean API contracts and a scalable component architecture to support rapid feature expansion.

Worklytics — HR & Payroll Intelligence Dashboard *HR & Payroll Intelligence Platform*

Stack: React.js · JavaScript (ES6+) · Vite · API Middleware · Authentication · RBAC · Git

- Built a full-stack HR analytics and payroll intelligence platform to monitor Saudization compliance, workforce KPIs, and hiring risk indicators across enterprise teams.
- Developed responsive React modules with reusable components and custom hooks, implementing token-based authentication with role-based access control (Admin, HR, Manager).
- Delivered core modules for Employees, Payroll, Attendance, Compliance, Approvals, and Reports with export-ready reporting workflows for regulatory submissions.

React Recon — Code Smell Detection Platform *Final Year Project (FYP)*

Stack: React.js · Node.js · Express · MongoDB (MERN) · Python · scikit-learn · Machine Learning

- Designed and built a full-stack web application that automatically detects code smells in public GitHub repositories using heuristic rules and a custom-trained machine learning model.
- Scraped, cleaned, and processed data from 300+ open-source repositories to construct a synthetic labeled dataset of 10,000+ code smell instances, achieving high ML model accuracy.
- Engineered a Python-based analysis engine integrated with the MERN stack backend, enabling on-demand repository analysis with results visualized through interactive dashboards.

EDUCATION

Bachelor of Science, Computer Science

National University of Computer and Emerging Sciences (FAST-NUCES) · 2019 – 2023 · Islamabad, Pakistan

- Graduated with a strong foundation in software engineering, data structures, algorithms, database systems, and computer networks.
- Final Year Project (React Recon) recognized for innovation in automated code quality analysis using ML.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3/SCSS

Frontend: Angular 17, React.js, Vite, RxJS, Redux, Context API, Tailwind CSS, Bootstrap, Chart.js, D3.js

Backend: Node.js, Express.js, RESTful API Design, JWT Authentication, WebSockets, GraphQL

Databases: MongoDB, MySQL, PostgreSQL, Mongoose ODM, Schema Design, Query Optimization

IoT & Platforms: Cumulocity IoT, MQTT Protocol, Sensor Integration, Real-Time Telemetry

DevOps & Tools: Git, GitHub, Docker (basic), Swagger/OpenAPI, Postman, JIRA, npm, Agile/Scrum

Testing: Unit Testing, REST API Testing, Cross-Browser Compatibility Testing

Practices: RBAC, SDLC, CI/CD, Code Reviews, Technical Documentation, MVC/MVVM

SOFT SKILLS

- Strong communicator and collaborator: experience in Agile/Scrum teams, cross-functional deployment squads, and client-facing project rollouts.
- Core strengths: Decision-Making · Problem-Solving · Team Leadership · Technical Documentation · Mentoring

References available upon request · Open to relocation and remote opportunities